

A TORSION-FREE NONSOFC HYPERBOLIC GROUP FROM MARKED LINES

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ABSTRACT. We present a candidate construction of a torsion-free nonsofic word-hyperbolic group with a finite two-dimensional classifying space. The starting point is the marked-line rank obstruction in OpenAI's family 252 construction of a non-residually finite hyperbolic group. We prove a quantitative extension of that obstruction to arbitrary finite permutation assignments: the rank deficit is bounded by the sum of the triangular Hamming defects and a weighted sum of the fixed-point proportions of rectangular words. The bound is uniform in the permutation degree and requires no stability theorem for approximate actions. A five-dimensional affine-line realization of the classical graph $D(5, q)$, together with a translated and linearly transformed interpolation grid, supplies the required incidence data. The associated triangle complex has a locally $\text{CAT}(-1)$ metric, and its rectangular words are nontrivial local geodesics. Consequently the permutation inequality excludes sofic approximations. In fact, one nonidentity element is killed by every homomorphism to every sofic group. The construction is specified by a terminating finite search; a complete presentation at the stated numerical parameters has not been materialized.

1. INTRODUCTION

Finite quotients detect an element of a group by means of an exact action on a finite set. Sofic approximations ask for less: on any prescribed finite part of the multiplication table, the action need only be correct at most points. An obstruction to residual finiteness therefore need not obstruct soficity. The distinction is decisive for constructions in which exact relations produce a linear-algebraic contradiction. To exclude soficity, the contradiction must survive small errors, uniformly in the size of the finite set.

The marked-line construction of OpenAI [6] is particularly suited to this question. It attaches triangular relations to a high-girth incidence graph. Those relations give low-rank factorizations, while certain four-edge loops force high rank in finite regular permutation representations. Polynomial interpolation combines the local factorizations into one global identity. Here we show that this identity also obstructs approximate permutation actions. The resulting geometric conclusion is the following.

Theorem 1.1. *There is a finite connected two-dimensional hyperbolic triangle complex K whose intrinsic metric is locally $\text{CAT}(-1)$ and whose fundamental group G is not sofic. In particular, G is torsion-free and word-hyperbolic, and K is a finite classifying space for G . Moreover, there is an element $w \neq 1$ of G such that every homomorphism from G to a sofic group sends w to the identity.*

All the data defining K can be specified by a finite deterministic procedure. An explicit finite choice is given in Proposition 3.5.

The main estimate is easy to describe before giving the construction. There are N marked lines, m points, and an exceptional set E . Each line carries an r -by- r array of edge permutations of degree D . Let δ_τ be the normalized Hamming defect of a triangular relation,

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and let λ_i be the average fixed-point proportion of the nondegenerate rectangles in the i th array. We prove

$$(1.1) \quad \frac{Nr}{4} - 2m - r|E| \leq \sum_{\tau} \delta_{\tau} + \frac{r^2}{8} \sum_{i=1}^N \lambda_i.$$

The data are chosen so that the left side is positive. The inequality holds for every positive permutation degree, even when D is divisible by the fixed prime q used in the proof. Since all the rectangle words are nonidentity in G , soficity would make every term on the right arbitrarily small.

The geometry and the algebra impose different requirements on the incidence data. Girth controls the links and makes the rectangles nontrivial. High degree outside a small exceptional set supplies the local factorizations. A grid of distinct marks supplies low-degree interpolation. We obtain these simultaneously from the classical Lazebnik–Ustimenko–Woldar graph $D(5, q)$ [5]. Its labels are actual affine lines. A perfect matching chooses one mark on each line, and a uniformly chosen affine image of a small grid selects a large subfamily with the needed degrees. Pairwise transitivity of the affine group is enough for the selection argument.

The original marked-line construction, triangular wiring, interpolation tensor identity, fixed-prime rank comparison, exact finite-quotient rank obstruction, and exceptional-set method are inherited from OpenAI family 252 [6, Section 2]. The contribution relative to that source is the quantitative inequality (1.1) for arbitrary approximate permutation assignments and its application to soficity. The five-dimensional selection and the compatible hyperbolic metric refine the geometric input. All ingredients used in the argument are reproduced below, apart from the cited standard facts about $\text{CAT}(-1)$ spaces. The graph $D(5, q)$ and the general $\text{CAT}(-1)$ machinery are classical.

Nonsocicity itself, even for finitely presented torsion-free groups, is already the subject of earlier constructions [8, 3, 4]. The issue addressed here is its coexistence with word hyperbolicity and a finite negatively curved two-dimensional classifying space. Chapman and Peled [2, Theorem 1.1] give a different route to nonsocic hyperbolic groups, conditional on a permutation cocycle-expansion estimate for random complexes. Our proof does not assume that estimate or any correction of approximate actions to exact ones. We do not assert an exhaustive priority claim.

We first formulate the finite rank obstruction and prove it. We then construct the incidence data, build the triangle complex, and verify its curvature and its rectangle words. Finally we pass from the finite obstruction to the group-theoretic conclusions. This order keeps the new quantitative step separate from its geometric input.

The proof has three steps. Rectangle freeness forces high rank via real trace estimates and reduction modulo a fixed prime. Approximate triangle relations force low-rank factorizations of local sums, charging at most $D\delta_{\tau}$ in rank for each Hamming defect. An exact interpolation identity combines these local estimates on one global matrix. The positive gap between its lower and upper rank bounds rules out sufficiently accurate sofic approximations.

2. THE FINITE RANK OBSTRUCTION

For $D \geq 1$ and $\sigma, \rho \in \text{Sym}(D)$, write

$$d_H(\sigma, \rho) = \frac{1}{D} |\{a \in \{1, \dots, D\} : \sigma(a) \neq \rho(a)\}|, \quad \text{fix}(\sigma) = \frac{|\text{Fix}(\sigma)|}{D}.$$

We identify permutations with their permutation matrices. Products of matrices use the same order as products of group elements, and the inverse matrix is the transpose. The metric d_H is bi-invariant, and $\text{fix}(\sigma) = 1 - d_H(\sigma, \text{id})$.

Fix an integer $r \geq 2$, a prime $q > r$, a finite point set P of cardinality m , and a nonempty label set I of cardinality N . For each point z , prescribe a subset $I(z) \subseteq I$. Also fix a proper

exceptional set $E \subset P$. At every $z \notin E$, suppose $I(z)$ is the disjoint union of two blocks; use no blocks at points of E . Index blocks at different points separately; write $p(b)$ for the point of a block. Each block is equipped with a bijection

$$(2.1) \quad I_b \longrightarrow (\mathbb{Z}/s_b\mathbb{Z}) \times (\mathbb{Z}/t_b\mathbb{Z}), \quad i \longmapsto (\alpha_b(i), \beta_b(i)), \quad s_b, t_b \geq r.$$

All additions to these coordinates are in their respective cyclic groups. We use $[r] = \{0, \dots, r-1\}$.

The last hypothesis is an interpolation certificate. For each $z \in P$, let $H_z \in M_N(\mathbb{F}_q)$ have rank at most one, and suppose

$$(2.2) \quad \sum_{z: i \in I(z)} H_z = -E_{ii} \quad (i \in I),$$

where E_{ii} is the diagonal matrix unit corresponding to i . Section 3 produces these matrices by interpolation at marked points. For the moment, only (2.2) matters.

Assign arbitrary permutations of D points to symbols

$$X_{iuv}, \quad L_{b,u,j}, \quad R_{b,v,k},$$

where $i \in I$, $u, v \in [r]$, $j \in \mathbb{Z}/s_b\mathbb{Z}$, and $k \in \mathbb{Z}/t_b\mathbb{Z}$. There is no assumption that this assignment extends to a group homomorphism. A triangle index is a tuple $\tau = (b, i, u, v)$ with $i \in I_b$. Define its defect by

$$(2.3) \quad \delta_{b,i,u,v} = d_H(L_{b,u,\alpha_b(i)+v} X_{iuv} R_{b,v,\beta_b(i)+u}^{-1}, \text{id}).$$

For each label, the rectangular products are

$$(2.4) \quad Q_{i;u,u',v,v'} = X_{iuv} X_{iu'v}^{-1} X_{iu'v'} X_{iuv'}^{-1}, \quad u \neq u', \quad v \neq v'.$$

Let λ_i be the average of $\text{fix}(Q_{i;u,u',v,v'})$ over these $r^2(r-1)^2$ ordered index tuples. Counting tuples, even when the corresponding words agree as group elements, is the convention throughout.

Notation. N : number of marked lines (labels); m : number of points; E : exceptional points; r : array size; D : permutation degree; δ_τ : triangle defect; λ_i : mean rectangle fixed-point proportion.

Theorem 2.1 (Quantitative marked-line obstruction). *Under the preceding hypotheses, every permutation assignment in every degree $D \geq 1$ satisfies*

$$(2.5) \quad \sum_{i \in I} \frac{r^2}{2(2r-1 + (r-1)^2 \lambda_i)} \leq 2(m - |E|) + r|E| + \sum_{\tau} \delta_{\tau}.$$

Consequently, if $\Delta = Nr/4 - 2m - r|E|$, then

$$(2.6) \quad \Delta \leq \sum_{\tau} \delta_{\tau} + \frac{r^2}{8} \sum_{i \in I} \lambda_i.$$

The prime and all the incidence data are fixed independently of D .

Lemma 2.2 transfers real rank to the fixed finite field; Lemma 2.3 obtains the lower bound from rectangle freeness; Lemma 2.4 gives the upper bound from defective triangle relations.

Lemma 2.2 (A fixed-prime rank comparison). *Let $r \geq 1$ and let M be an integer matrix with row norms at most $\sqrt{r}$. For every prime $q > r$,*

$$\text{rank}_{\mathbb{F}_q} M \geq \frac{1}{2} \text{rank}_{\mathbb{R}} M.$$

Proof. Put $d = \text{rank}_{\mathbb{R}} M$. If $d > 0$, choose a nonsingular d -by- d minor B . Hadamard's inequality gives $1 \leq |\det B| \leq r^{d/2}$. Let d' be its rank modulo q . Choose d' rows spanning its row space modulo q . Subtracting integer lifts of linear combinations of those rows from the remaining rows preserves the determinant and makes $d - d'$ rows divisible by q . Thus

$$q^{d-d'} \mid \det B, \quad q^{d-d'} \leq r^{d/2}.$$

Since $q > r$, this implies $d' \geq d/2$. The rank of M modulo q is at least d' . The case $d = 0$ is immediate. $\square$

This is the fixed-prime comparison used in [6]. The permutation degree will enter only as a dimension in inequalities over the real numbers. No division by D will take place in $\mathbb{F}_q$.

Lemma 2.3 (Rectangles force rank). *For a label i , let $Y_i = (X_{iuv})_{u,v \in [r]}$, an rD -by- rD integer block matrix. Then*

$$(2.7) \quad \text{rank}_{\mathbb{F}_q} Y_i \geq \frac{r^2 D}{2(2r-1 + (r-1)^2 \lambda_i)} \geq D \left(\frac{r}{4} - \frac{r^2 \lambda_i}{8} \right).$$

Proof. The first two trace moments of $Y_i Y_i^\top$ have a direct permutation interpretation. The first is $r^2 D$. In the second, each term is the trace of the rectangular product (2.4), allowing degenerate indices. There are $2r^3 - r^2$ tuples with $u = u'$ or $v = v'$, and their products are the identity. The other $r^2(r-1)^2$ terms have average trace $D\lambda_i$. Hence

$$\begin{aligned} \text{tr}(Y_i Y_i^\top) &= r^2 D, \\ \text{tr}((Y_i Y_i^\top)^2) &= r^2 D(2r-1 + (r-1)^2 \lambda_i). \end{aligned}$$

Cauchy-Schwarz applied to the nonzero eigenvalues of this positive semidefinite matrix gives

$$\text{rank}_{\mathbb{R}} Y_i \geq \frac{r^2 D}{2r-1 + (r-1)^2 \lambda_i}.$$

Every row of Y_i has exactly r unit entries in distinct block columns. Lemma 2.2 proves the first inequality.

For the second, put $A = 2r-1$ and $B = (r-1)^2$. For $\lambda \geq 0$,

$$\frac{r^2}{2A} - \frac{r^2}{2(A+B\lambda)} = \frac{r^2 B \lambda}{2A(A+B\lambda)} \leq \frac{r^2 \lambda}{8},$$

because $B/A^2 \leq 1/4$. Also $r^2/(2A) \geq r/4$. No smallness assumption on λ is needed. $\square$

Lemma 2.4 (The cost of triangular errors). *Over $\mathbb{F}_q$, put $Z_z = \sum_{i \in I(z)} Y_i$. For $z \notin E$,*

$$(2.8) \quad \text{rank } Z_z \leq 2D + D \sum_{\tau: p(b)=z} \delta_\tau.$$

For $z \in E$, one has $\text{rank } Z_z \leq rD$.

Proof. Two permutation matrices that disagree on k points differ in at most k columns. Their difference therefore has rank at most k over any field. By bi-invariance of d_H , the error matrix

$$A_{b,i,u,v} = X_{iuv} - L_{b,u,\alpha_b(i)+v}^{-1} R_{b,v,\beta_b(i)+u}$$

has rank at most $D\delta_{b,i,u,v}$ over $\mathbb{F}_q$.

Fix one block b and replace every X_{iuv} in $\sum_{i \in I_b} Y_i$ by the displayed ideal product. For each fixed u, v , the shifted coordinates $(\alpha_b(i) + v, \beta_b(i) + u)$ run through the full Cartesian product exactly once. The resulting block matrix is therefore

$$(2.9) \quad \left(\sum_{j \in \mathbb{Z}/s_b \mathbb{Z}} L_{b,u,j}^{-1} \right)_{u \in [r]} \left(\sum_{k \in \mathbb{Z}/t_b \mathbb{Z}} R_{b,v,k} \right)_{v \in [r]}.$$

The first factor is an rD -by- D block column and the second a D -by- rD block row. Its rank is at most D . In particular, the factorization concerns the entire array, not just its separate entries. The order of matrix multiplication is unchanged.

Embedding an error $A_{b,i,u,v}$ into its (u, v) block does not increase its rank. Rank subadditivity bounds the cost of restoring all entries by $D \sum_{i,u,v} \delta_{b,i,u,v}$. There are two blocks at a regular point, proving (2.8). At an exceptional point the ambient matrix size gives the asserted bound. Its actual incidences remain in Z_z . $\square$

Proof of Theorem 2.1. The certificate (2.2) gives the identity

$$(2.10) \quad -\bigoplus_{i \in I} Y_i = \sum_{z \in P} H_z \otimes Z_z.$$

Indeed, expanding Z_z and exchanging the two finite sums makes the coefficient of Y_i equal to $-E_{ii}$. Both sides are NrD -by- NrD matrices. This identity is independent of the triangular errors.

Ranks add on the left. On the right, $\text{rank}(H_z \otimes Z_z) \leq \text{rank } Z_z$, since $\text{rank } H_z \leq 1$. Lemma 2.4 and rank subadditivity yield

$$\sum_{i \in I} \text{rank}_{\mathbb{F}_q} Y_i \leq D \left(2(m - |E|) + r|E| + \sum_{\tau} \delta_{\tau} \right).$$

Insert the first bound of Lemma 2.3 and cancel the positive real number D to obtain (2.5). The second bound of that lemma, and the harmless weakening $2(m - |E|) \leq 2m$, give (2.6). $\square$

Let $S_{\text{reg}} = \sum_{z \notin E} |I(z)|$. The number of triangle indices is $T = r^2 S_{\text{reg}}$. If $\bar{\delta}$ is their mean defect and $\bar{\lambda} = N^{-1} \sum_i \lambda_i$, the estimate reads

$$(2.11) \quad \Delta \leq T\bar{\delta} + \frac{Nr^2}{8} \bar{\lambda}.$$

In particular, if $\Delta > 0$ and $T > 0$, every assignment has

$$\max\{\bar{\delta}, \bar{\lambda}\} \geq \frac{\Delta}{T + Nr^2/8} > 0.$$

This is the quantitative obstruction needed below. The stronger nonlinear estimate (2.5) is retained because it makes the dependence on the rectangle errors explicit.

3. MARKED AFFINE LINES IN FIVE DIMENSIONS

We now construct data satisfying the preceding hypotheses, with positive Δ and incidence girth at least ten. This is a particular realization of the general obstruction, which assumes neither affine lines nor dimension five. The graph used here is $D(5, q)$ of Lazebnik, Ustimenko and Woldar [5, Section 2, Proposition 2.1]. We give its coordinates and the short girth argument, since the fact that its labels are affine lines is also needed for the algebra.

Take two copies of $\mathbb{F}_q^5$. Write point coordinates as (t, y, z, w, v) and label coordinates as $[a, b, c, d, e]$. Incidence means

$$(3.1) \quad b - y = at, \quad c - z = bt, \quad d - w = ay, \quad e - v = az.$$

For a fixed label, its incident points form the full affine line

$$(3.2) \quad (t, b - at, c - bt, d - ab + a^2t, e - ac + abt), \quad t \in \mathbb{F}_q.$$

At a fixed point, specifying a determines a unique incident label. Thus the graph is simple and q -regular on each side, with q^5 vertices on each side. Distinct labels give distinct affine lines: the normalized direction recovers a, b , and the point with $t = 0$ then recovers c, d, e .

Lemma 3.1. *The graph defined by (3.1) has girth at least ten.*

Proof. Consider a simple cycle with s label vertices. In cyclic order, write the first two coordinates of its labels as a_j, b_j , and let $\tau_j \neq 0$ be the change in the first point coordinate along the j th line. Adjacent a_j are distinct, including across the closing seam, because a point lies on only one label with a given first coordinate. Closing the cycle in all five coordinates gives

$$(3.3) \quad \sum_j \tau_j = \sum_j a_j \tau_j = \sum_j a_j^2 \tau_j = \sum_j b_j \tau_j = \sum_j a_j b_j \tau_j = 0.$$

There is no four-cycle, since two distinct affine lines cannot share two distinct points. If $s = 3$, the three a_j are distinct, and the first three equations form an invertible Vandermonde system. This would force all τ_j to vanish.

Suppose $s = 4$. There are at least two distinct a -values. If there are exactly three, combine the coefficients τ_j having equal a_j . The three-value Vandermonde system makes all three combined coefficients zero, including the coefficients of the two singleton values. This again contradicts $\tau_j \neq 0$.

If all four a_j are distinct, the rows (τ_j) and $(a_j\tau_j)$ are independent. Their common kernel has dimension two. By the first three equations of (3.3), the independent vectors $(1)_j$ and $(a_j)_j$ belong to this kernel. The last two equations therefore imply $b_j = A + Ba_j$ for some $A, B \in \mathbb{F}_q$.

It remains to consider two a -values. They alternate around the cycle. The first two equations give $\tau_3 = -\tau_1$ and $\tau_4 = -\tau_2$. The last two, using the two distinct a -values, give $b_3 = b_1$ and $b_4 = b_2$. Thus $b_j = A + Ba_j$ in this case also.

In either of these last cases, the intersection of adjacent labels has first coordinate determined by

$$b_{j+1} - b_j = (a_{j+1} - a_j)t.$$

It follows that every such intersection has $t = B$. Both endpoints of every line segment in the cycle then have the same first coordinate, contrary to $\tau_j \neq 0$. This excludes eight-cycles and completes the proof. The argument is valid in every characteristic. $\square$

There is a useful explicit perfect matching in this graph. Match a point to its incident label with $a = t$. Conversely, match a label to its unique point with $t = a$. These operations are inverse. In particular, every label has a distinct matched point on its line.

Let $S \subset \mathbb{F}_q$ have cardinality s , and let $B = U(S^5)$, where $U \in \text{AGL}(5, q)$ is an affine bijection. Retain exactly the labels whose matched points belong to B , and use those matched points as marks. Index the retained labels by I , so $N = s^5$. If a_i is the mark of label i , define

$$I(x) = \{i \in I : x \in \ell_i \setminus \{a_i\}\}.$$

Only a line's own mark is removed from its incidence set. Any other marks lying on that line are retained as incident points. The punctured incidence graph is a subgraph of $D(5, q)$ and hence still has girth at least ten.

Lemma 3.2 (Selecting the grid). *Put $m = q^5$, $N = s^5$, and $\rho = N/m$. Choose U uniformly in $\text{AGL}(5, q)$. The punctured degree $d_x = |I(x)|$ at every fixed point has mean $\mu = (q-1)\rho$ and variance at most μ . If $\mu \geq 8r^2$ and $E = \{x : d_x < 4r^2\}$, then*

$$(3.4) \quad \Pr(|E| > m/(2r)) \leq \frac{8r}{\mu}.$$

Proof. At a fixed point x , the q incident labels have q distinct matched points. Exactly one is x . Let A_x be the other $q-1$ matched points. The puncturing convention gives the exact identity

$$d_x = |A_x \cap U(S^5)|.$$

The set A_x is fixed before U is chosen.

The affine group is transitive on ordered pairs of distinct points. For distinct $a, a' \in P$ this gives

$$\Pr(a \in B) = \rho, \quad \Pr(a, a' \in B) = \frac{N(N-1)}{m(m-1)}.$$

The covariance of the two membership indicators is consequently $-\rho(1-\rho)/(m-1)$. Summing over A_x proves

$$\mathbb{E}d_x = \mu, \quad \text{Var}(d_x) = (q-1)\rho(1-\rho) \left(1 - \frac{q-2}{m-1}\right) \leq \mu.$$

If $\mu \geq 8r^2$, a degree below $4r^2$ differs from its mean by at least $\mu/2$. Chebyshev's inequality gives $\Pr(x \in E) \leq 4/\mu$. Summing over x gives $\mathbb{E}|E| \leq 4m/\mu$, and Markov's inequality proves (3.4). Independence of different degrees is not required. $\square$

The affine map need not preserve the incidence graph. It moves the grid of marks inside the fixed point space; the retained labels remain the original affine lines. This distinction allows the same choice of marks to satisfy both the degree estimate and the interpolation requirement.

Lemma 3.3 (Interpolation on punctured lines). *If $10(s-1) < q-1$, the retained marked lines admit matrices H_x of rank at most one satisfying (2.2).*

Proof. For $c \in S$, let

$$p_c(T) = \prod_{d \in S \setminus \{c\}} \frac{T-d}{c-d}.$$

Write $U^{-1}(a_i) = (c_{i1}, \dots, c_{i5})$, and define

$$f_i(X) = \prod_{j=1}^5 p_{c_{ij}}((U^{-1}X)_j).$$

These polynomials have total degree at most $5(s-1)$ and satisfy $f_i(a_{i'}) = \mathbf{1}_{i=i'}$. Let $f = (f_i)_{i \in I}$ be a column vector and put $H_x = f(x)f(x)^\top$. Thus $\text{rank } H_x \leq 1$ and $H_{a_i} = E_{ii}$.

Every entry of H has degree at most $10(s-1)$ on an affine line. A polynomial of degree less than $q-1$ sums to zero over $\mathbb{F}_q$. For the constant term the sum is $q=0$ in the field. For a monomial T^j with $0 < j < q-1$, choose $a \in \mathbb{F}_q^\times$ with $a^j \neq 1$; multiplication by a permutes $\mathbb{F}_q$ and forces its power sum to vanish. Applying this entrywise on the full line ℓ_i gives

$$\sum_{x \in \ell_i \setminus \{a_i\}} H_x = -H_{a_i} = -E_{ii}.$$

This is exactly (2.2). $\square$

High degree supplies the two Cartesian blocks without further divisibility assumptions.

Lemma 3.4. *Every integer $d \geq 4r^2$ can be written as the sum of two products $s_1 t_1 + s_2 t_2$ in which all four factors are at least r .*

Proof. Choose $n_2 \in \{r, \dots, 2r-1\}$ congruent to d modulo r . Then

$$d = rn_1 + (r+1)n_2, \quad n_1 = \frac{d - (r+1)n_2}{r} \geq \frac{2r^2 - r + 1}{r} \geq r.$$

The products rn_1 and $(r+1)n_2$ have the required form. $\square$

Proposition 3.5 (A numerical instance). *Take $q = 2^{61} - 1$, $r = 1\,200\,000$, $s = \lfloor q/10 \rfloor$, and $S = \{0, \dots, s-1\} \subset \mathbb{F}_q$. There is an affine bijection U for which the preceding construction has $|E| \leq m/(2r)$ and*

$$(3.5) \quad N = s^5, \quad m = q^5, \quad \Delta = Nr/4 - 2m - r|E| \geq m/5.$$

It satisfies all the hypotheses of Theorem 2.1, and its incidence graph has girth at least ten.

Proof. The number $2^{61} - 1$ is prime. This familiar Mersenne prime can also be certified by the Lucas–Lehmer recurrence: with $u_0 = 4$ and $u_{j+1} = u_j^2 - 2$ modulo $2^{61} - 1$, one has $u_{59} = 0$; the exponent 61 is prime. The explicit prime is only a convenient instance; the following inequalities also hold for all sufficiently large primes with r fixed.

Put $c_0 = 10^{-5}$. Direct integer comparison gives

$$q \geq \frac{16r^2}{c_0}, \quad q > r.$$

Since $s \geq q/10 - 1$ and $q > 500$, Bernoulli's inequality implies

$$c_0(1 - 10/q)^5 \leq \frac{N}{m} \leq c_0, \quad \frac{N}{m} \geq 0.9c_0.$$

Consequently

$$\mu = (q - 1)N/m \geq c_0q/2 \geq 8r^2.$$

Lemma 3.2 shows that the desired bound on $|E|$ holds with probability at least $1 - 8r/\mu \geq 1 - 1/r > 0$. Choose such a U . Since $c_0r = 12$,

$$\Delta \geq \left(\frac{0.9c_0r}{4} - 2 - \frac{1}{2} \right) m = \frac{m}{5}.$$

At every point outside E , apply Lemma 3.4 and choose arbitrary bijections (2.1). Lemma 3.1 gives the girth bound, and $10(s - 1) < q - 1$ permits Lemma 3.3. $\square$

For definiteness, order field elements by their representatives in $\{0, \dots, q - 1\}$, order the coefficients of affine maps lexicographically, and choose the first invertible affine map for which $|E| \leq m/(2r)$. Membership in E is determined by finite enumeration of the incidences. The proposition proves that this search terminates. Ordering labels and filling the two Cartesian blocks in lexicographic order then removes all further choices. This specifies one finite complex by a terminating search; it does not provide a practical computation of its full incidence table or presentation.

4. THE TRIANGLE COMPLEX AND ITS CURVATURE

Fix the data of Proposition 3.5. Following the wiring in [6], take three distinct vertices O, V, W and oriented edges

$$\begin{aligned} x_{iuv} : V &\longrightarrow W & (i \in I, u, v \in [r]), \\ L_{b,u,j} : O &\longrightarrow V & (u \in [r], j \in \mathbb{Z}/s_b\mathbb{Z}), \\ R_{b,v,k} : O &\longrightarrow W & (v \in [r], k \in \mathbb{Z}/t_b\mathbb{Z}). \end{aligned}$$

Different indices denote different edges. For every triangle index $\tau = (b, i, u, v)$, attach a triangle whose boundary, read in path order, is

$$(4.1) \quad L_{b,u,\alpha_b(i)+v} x_{iuv} R_{b,v,\beta_b(i)+u}^{-1}.$$

Each closed cell embeds: its vertices O, V, W and its three edges are distinct. There are regular points, so there are L - and R -edges and the complex is connected. Denote it by K .

Give each face the hyperbolic triangle with angles

$$(4.2) \quad \theta_O = \frac{21\pi}{40}, \quad \theta_V = \theta_W = \frac{9\pi}{40}.$$

The angle sum is $39\pi/40 < \pi$. All L - and R -edges have the corresponding equal side length; all x -edges have the third side length, including any unused ones. Hence all gluings are isometric.

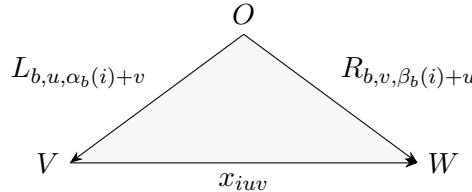


FIGURE 1. The incidence of the three oriented sides of one face. The drawing is schematic; the metric face is the hyperbolic triangle with angles (4.2).

Recall that the angular link at a vertex has a vertex for each incident edge germ and an edge for each face corner, with length equal to that corner's angle. A reduced path has

no immediate edge reversal. Reduction is also required at the closing seam of a cyclically reduced closed path. Disconnected directions have infinite link distance.

Lemma 4.1. *Every nonempty cyclically reduced closed path in the link of O has length at least $21\pi/10$. At V and W the corresponding bound is $9\pi/4$. Moreover, at either V or W , distinct x -directions with the same label i have link distance at least $9\pi/4$, or lie in different components.*

Proof. At O , a block contributes a complete bipartite graph between its L - and R -directions. To see that it is simple, fix $L_{b,u,j}$ and $R_{b,v,k}$. Their corner must use the unique label with

$$\alpha_b(i) = j - v, \quad \beta_b(i) = k - u.$$

Directions belonging to different blocks are distinct. Thus the whole link is a disjoint union of simple complete bipartite graphs. Its nonempty reduced closed paths have at least four edges, with length at least $4\theta_O = 21\pi/10$.

Map the link at V to the punctured incidence graph by sending the direction x_{iuv} to i and the direction $L_{b,u,j}$ to $p(b)$. Each corner maps to the corresponding incidence edge. This is a graph immersion. At an x_{iuv} -direction, a specified point z allows at most one corner, since i belongs to at most one block at z . At an $L_{b,u,j}$ -direction, a specified label i forces $v = j - \alpha_b(i)$ modulo s_b ; there is at most one such $v \in [r]$ because $s_b \geq r$. This proves local injectivity on incident edges. The argument at W is identical, using $u = k - \beta_b(i)$ modulo t_b and $t_b \geq r$.

An immersion preserves reduction, including reduction at a closing seam. Every nonempty cyclically reduced closed link path therefore maps to a reduced closed walk in the incidence graph. Such a walk contains a cycle. By Lemma 3.1, it has at least ten edges, and its angular length is at least $10\theta_V = 9\pi/4$.

Finally, a reduced path between distinct x -directions with the same label maps to a nonempty closed walk reduced at every internal vertex. The closing seam need not be reduced. Nevertheless the walk contains a simple cycle: a shortest positive subinterval with equal endpoints cannot have length one in a simple graph, or length two without an internal reversal. It is a simple cycle of length at least ten. Applying this to a shortest link path proves the distance assertion. $\square$

Proposition 4.2. *The universal cover of K is a proper $\text{CAT}(-1)$ space. The group $G = \pi_1(K, O)$ acts on it freely, properly and cocompactly by isometries. It follows that G is torsion-free and word-hyperbolic and that K is a finite two-dimensional $K(G, 1)$.*

Proof. All closed reduced link paths in Lemma 4.1 have length greater than 2π . The metric link criterion therefore makes K locally $\text{CAT}(-1)$; see [1, Theorem II.5.2 and Lemma II.5.6]. There are finitely many shapes. The interiors of edges and faces satisfy the same local curvature condition.

For a convention requiring intersections of cells to be single faces, take a compatible geodesic barycentric subdivision. The closed cells embed, so distinct cells provide distinct barycentric vertices, even when there are parallel original edges. This produces a genuine finite simplicial subdivision with the same metric. At a new face-interior vertex the link is a circle of length 2π ; at a new edge-interior vertex it is a union of length- π paths between the two edge directions. These also satisfy the link criterion.

The lifted metric on the universal cover is complete. Cartan–Hadamard [1, Theorem II.4.1] makes it $\text{CAT}(-1)$. A cover of a finite complex is locally finite, and with these finitely many positive cell shapes the complete length metric is proper. Deck transformations act freely and properly, and the quotient is the compact complex K .

A finite group of isometries of a complete $\text{CAT}(0)$ space fixes a point [1, Corollary II.2.8]. A finite-order deck transformation must therefore be the identity, so G is torsion-free. A $\text{CAT}(-1)$ space is Gromov-hyperbolic, and a proper cocompact isometric action on this geodesic space identifies G , up to quasi-isometry, with the universal cover [1, Chapters I.8

and III.H]. Hence G is word-hyperbolic. Finally, a $\text{CAT}(-1)$ space is contractible, which proves the classifying-space assertion. $\square$

Choose a path c_A from O to each vertex A , with c_O empty. For an oriented edge $e : A \rightarrow B$, let g_e be the class of $c_A e c_B^{-1}$. We multiply loops in path order. Each face gives

$$(4.3) \quad g_{L_{b,u,\alpha_b(i)+v}} g_{x_{iu}} g_{R_{b,v,\beta_b(i)+u}}^{-1} = 1.$$

Choosing the c_A in a spanning tree gives the usual finite presentation: the edge symbols, the triangular relators, and the relations setting tree edges equal to the identity.

Lemma 4.3. *Every element*

$$(4.4) \quad w_{i;u,u',v,v'} = g_{x_{iu}} g_{x_{iu'}^{-1}} g_{x_{iv}} g_{x_{iv'}^{-1}}, \quad u \neq u', \quad v \neq v',$$

is nonidentity in G .

Proof. After cancellation of the connecting paths, the word is conjugate to the closed four-edge path $x_{iu} x_{iu'}^{-1} x_{iv} x_{iv'}^{-1}$. Every turn compares distinct x -directions with the same label. Their link distance is greater than π by Lemma 4.1, so the path is a positive-length closed local geodesic. If it were nullhomotopic, its lift to the universal cover would be a positive-length closed local geodesic. In a $\text{CAT}(-1)$ space every local geodesic is a geodesic, which rules this out [1, Proposition II.1.4]. $\square$

5. EXCLUDING SOFC APPROXIMATIONS

We use the usual finite-set definition of soficity [9]. A group H is sofic if, for every finite $F \subset H$ containing 1 and every $\eta > 0$, there exist $D \geq 1$ and a map $\phi : F \rightarrow \text{Sym}(D)$ such that $\phi(1) = \text{id}$,

$$d_H(\phi(a)\phi(b), \phi(ab)) < \eta \quad (a, b, ab \in F),$$

and $d_H(\phi(a), \text{id}) > 1 - \eta$ for $a \in F \setminus \{1\}$. This version includes asymptotic freeness as part of the definition.

We spell out the finite-word passage to keep the quantifiers clear. Fix finitely many words in a finite alphabet of elements of H and their inverses. Include in F the letters, their inverses, all word prefixes, and 1. If a nonempty word has length ℓ and has k inverse letters, evaluating it in the permutations assigned to the positive letters differs from ϕ of its value by at most $(\ell - 1 + k)\eta$. Indeed, each inverse substitution costs less than η , by approximate multiplication of a and a^{-1} , and each prefix multiplication costs less than η .

Proposition 5.1. *For the group G constructed above, there is no homomorphism $f : G \rightarrow H$ to a sofic group for which every rectangle (4.4) has nonidentity image.*

Proof. Suppose such an f exists. Apply the finite-word observation in H to all the images of the triangular words and the rectangular words, using the edge elements as the positive alphabet. There are finitely many of each, with the incidence data already fixed. Choose a sofic model of the resulting finite set and assign $\phi(f(g_e))$ to every edge e .

Each triangular word has length three and one inverse letter. Its group value is 1 by (4.3), so its Hamming defect is less than 3η . A rectangular word has length four and two inverse letters. Its evaluated permutation is within 5η of $\phi(f(w))$. Since $f(w) \neq 1$, the latter has fixed-point proportion less than η . The fixed-point proportion of the evaluated rectangle is therefore less than 6η . These estimates also cover edge elements that happen to be the identity.

To obtain an explicit contradiction, set

$$(5.1) \quad \varepsilon_0 = \frac{1}{100rq}.$$

Choose $\eta < \varepsilon_0/6$. Then every triangle defect and every rectangle fixed-point proportion is less than ε_0 . Each retained line has exactly $q - 1$ punctured incidences, so $S_{\text{reg}} \leq N(q - 1)$. Theorem 2.1 would imply

$$\begin{aligned} \Delta &\leq r^2(S_{\text{reg}} + N/8)\varepsilon_0 \\ &\leq r^2 N q \varepsilon_0 = \frac{rN}{100} \leq \frac{10^{-5}r}{100} m = \frac{3m}{25}. \end{aligned}$$

This contradicts $\Delta \geq m/5$ from Proposition 3.5. $\square$

Proof of Theorem 1.1. Proposition 4.2 gives all the geometric conclusions, and Lemma 4.3 shows that every rectangle is nonidentity. If G were sofic, its identity homomorphism would contradict Proposition 5.1.

For the last assertion, let $\mathcal{W}$ be the finite set of rectangle elements. Suppose that for every $w \in \mathcal{W}$ there were a homomorphism $f_w : G \rightarrow H_w$ to a sofic group with $f_w(w) \neq 1$. The product map into $\prod_{w \in \mathcal{W}} H_w$ would preserve every rectangle. Finite products of sofic groups are sofic: for a finite set in a product of k factors and a tolerance $\eta > 0$, approximate its coordinate projections with error less than η/k and act on the Cartesian product of the finite sets. Multiplication errors add at most, while fixed-point proportions multiply; a nonidentity coordinate supplies freeness. This contradicts Proposition 5.1. Hence some fixed $w \in \mathcal{W}$ is killed by all homomorphisms to sofic groups, and $w \neq 1$ by Lemma 4.3. $\square$

The final argument identifies a finite set containing such a w ; it does not identify which rectangle has the universal kernel property. This quantifier distinction is useful: for each target there is a killed rectangle already by Proposition 5.1, and closure under finite products is what produces one rectangle working for all targets.

6. RELATION TO THE SOURCE CONSTRUCTION AND FURTHER SCOPE

The quantitative argument also applies directly to the original family 252 incidence data. That source uses marked lines in $\mathbb{F}_q^{20}$, a punctured incidence graph of girth at least fourteen, the same two-block partitions, and the strict inequality $Nr/4 > 2m + r|E|$. Its interpolation matrices satisfy (2.2). Theorem 2.1 therefore applies without any change in the combinatorial presentation. One may replace its Euclidean face metric by the common hyperbolic shape (4.2); girth fourteen is more than sufficient for Lemma 4.1. Thus every instance satisfying the stated source hypotheses has the same obstruction to soficity. The five-dimensional construction gives a different, smaller input to the same mechanism.

This also explains precisely where the passage beyond residual finiteness occurs. In [6], nontrivial rectangles act without fixed points in a regular representation of a finite quotient. Here their traces may be positive, and the triangular factorizations may fail. Lemmas 2.3 and 2.4 measure both losses. The tensor identity does not require modification. The advertised formalization scope of family 252 is its non-residual-finiteness theorem [7]; it does not certify the approximate-permutation extension proved here.

For comparison with the literature, OpenAI's earlier construction [8, Chapter 3] gives a nonsofic group through a property-(T) and expander mechanism. Fournier-Facio [3, Theorem 1.3] gives a finitely presented torsion-free nonsofic group, and Kun–Thom [4, Theorem A] give nonsofic generalized wreath products and doubles. Their discussion of a hyperbolic group admitting a nonsofic probability-measure-preserving action is a different assertion from nonsoficity of the group itself. The conditional hyperbolic construction of Chapman–Peled [2] uses permutation cocycle stability in a random complex. The rank estimate here instead acts directly on the approximate data. None of these comparisons is needed as an input to the proof.

The size reduction does not make the numerical construction small. For the parameters of Proposition 3.5, $m = q^5$ has 92 decimal digits, $N = \lfloor q/10 \rfloor^5$ has 87, and the bound $r^2 N(q - 1)$ for the number of triangles has 118. An affine map U has only thirty field

coefficients, but the present acceptance test evaluates a huge incidence ledger. A short symbolic certificate for a particular admissible U would make the construction easier to inspect. Existence of an admissible U , however, follows from the finite counting argument above and does not depend on actually running that search.

The obstruction is specific to permutations. It gives no nonhyperlinearity conclusion: a unitary matrix $e^{it}I_D$ can approach I_D in normalized Hilbert–Schmidt norm while their difference has rank D . Nor does rectangle separation in normalized rank replace Lemma 2.3 for arbitrary invertible matrices. For distinct positive real numbers a_u, b_v , the blocks $X_{uv} = (a_u + b_v)I_D$ form a matrix of rank $2D$, while every nondegenerate rectangle differs from the identity by a full-rank matrix. Thus neither nonhyperlinearity nor failure of linear softicity is established by the present argument. No property- (T) conclusion is asserted for G .

Disclosure. The mathematical derivation and manuscript were produced with substantial assistance from OpenAI language models. The manuscript is released as a candidate proof for independent mathematical scrutiny and has not undergone external peer review. The human contact listed above has not independently certified every argument. The approximate-permutation extension and the five-dimensional construction have not been formally verified.

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